Contexts and pseudocontexts

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In the projection lattice of a finite-dimensional Hilbert space, an orthogonal basis—a *context*—is characterised by the property that its atomic probabilities sum to one on every state. We study a weakening of this condition in which two tuples of atoms induce the same probability sum on every state. Such pairs are called *pseudocontexts*. In finite dimension this is equivalent to two distinct rank-one decompositions of the same positive operator, placing the notion naturally among finite frame-theoretic constructions. Size $k=1$ is trivial; size $k=2$ already admits genuinely nonorthogonal examples over $\mathbb{C}$; and, in fact, examples exist for every $k \geq 2$. For $k \geq 3$, genuine pseudocontexts also exist over $\mathbb{R}$. We present the operator formulation, give a general explicit construction, analyse the spectral structure of the associated structure operator, illustrate the symmetric $k=3$ case with a figure, and list the remaining classification problems.

Keywords: projection lattice, pseudocontext, rank-one decomposition, structure operator, quantum contextuality

I. INTRODUCTION

Let L be the lattice of projections in a Hilbert space of finite dimension $n \geq 2$. Its atoms (minimal non-zero elements) correspond to one-dimensional subspaces, represented by nonzero vectors. States are described by probability measures $P\colon L \to [0,1]$. We denote by $S(L)$ the set of all states on L.

Vectors $a_1, \ldots, a_n$ form a *context* if they compose an orthogonal basis. In this case (and only in this case) they satisfy

$$\forall P \in S(L): \quad \sum_{i=1}^{n} P(a_i) = 1. \tag{1}$$

The present note concerns weaker additive identifications obtained by comparing two tuples of atoms that are not required to be orthogonal. Such *pseudocontexts* are quantum-logical shadows of orthomodular-lattice constructions of Mayet–Navara–Rogalewicz [1] and have direct physical meaning: they encode state-independent empirical equivalences between families of generally noncommuting observables. It is then natural to ask for which sizes k and over which scalar fields genuine pseudocontexts exist. The answer is simple: pseudocontexts already exist for $k=2$ over $\mathbb{C}$, and, more generally, for every $k \geq 2$; for $k \geq 3$ they exist over $\mathbb{R}$ as well.

II. PRELIMINARIES

Throughout, H is a Hilbert space of finite dimension $n \geq 2$, and $L = L(H)$ is the orthomodular lattice of its closed subspaces. Its atoms correspond to one-dimensional subspaces, or equivalently to *rays* spanned by nonzero vectors. We identify an atom with any representative unit vector a and write $P_a = |a\rangle\langle a|$ for the orthogonal projection onto the ray it spans. More generally, for nonzero a we set $P_a = |a\rangle\langle a| / \langle a, a\rangle$, so that P_a depends only on the ray $\mathbb{C}a$.

A *state* is a map $P\colon L \to [0,1]$ with $P(I) = 1$ that is finitely additive on orthogonal families. In dimension $n \geq 3$, Gleason's theorem [2] asserts that every state arises from a density operator ρ via

$$P(a) = \operatorname{tr}(\rho P_a), \qquad \rho \geq 0, \quad \operatorname{tr}\rho = 1. \tag{2}$$

For $n=2$, Gleason's theorem fails for the abstract definition of a state, and we adopt (2) as the definition: a state is the assignment $P_a \mapsto \operatorname{tr}(\rho P_a)$ for some density operator ρ. Under either reading, the key consequence we use repeatedly is that two self-adjoint operators that agree on every state are equal as operators (the density operators in finite dimension separate self-adjoint operators).

III. CONTEXTS AND PSEUDOCONTEXTS

The operator form of (1) is elementary and is the starting point for everything that follows.

Proposition 1 (Contexts)**.** *Unit vectors $a_1, \ldots, a_n \in H$ form an orthonormal basis if and only if*

$$\sum_{i=1}^{n} P_{a_i} = I. \tag{3}$$

Proof. The forward implication is immediate. For the converse, fix j and evaluate (3) on the vector a_j:

$$1 = \langle a_j, I a_j\rangle = \sum_{i=1}^{n} |\langle a_i, a_j\rangle|^2.$$

The term with $i = j$ already contributes 1, so all others must vanish. Hence $\langle a_i, a_j\rangle = 0$ for $i \neq j$. As this holds for every j, the family is orthonormal, and being of size n in an n-dimensional space it is an orthonormal basis. $\square$

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Combined with (2), Proposition 1 recovers (1) as a state-theoretic characterisation of orthogonal bases.

Definition 2 (Pseudocontext)**.** Let $k \geq 1$. A *pseudocontext of size* k is a pair of k-tuples of atoms $(a_1,\dots,a_k)$, $(b_1,\dots,b_k)$ such that

(i) each tuple contains pairwise distinct atoms (i.e. no two of the a_i are scalar multiples, and likewise for the b_i),

(ii) the two tuples are disjoint as sets of rays, so that $|\{a_1,\dots,a_k,b_1,\dots,b_k\}| = 2k$, and

(iii) $\forall P \in S(L): \quad \sum_{i=1}^{k} P(a_i) = \sum_{i=1}^{k} P(b_i).$

A pseudocontext is called *genuine* if neither k-tuple is a context.

Remark 3. Condition (i) excludes repeated atoms within a single tuple, and condition (ii) ensures the two tuples are truly distinct as collections of rays. Without condition (ii), every pseudocontext of size k would yield pseudocontexts of every size $k' > k$ by adjoining $k' - k$ shared atoms; this trivial inflation can be carried out within the original ambient subspace. Condition (ii) rules out such padding and makes the size k a genuine invariant of the construction.

By (2), condition (iii) of Definition 2 is equivalent to the operator identity

$$T := \sum_{i=1}^{k} P_{a_i} = \sum_{i=1}^{k} P_{b_i}, \tag{4}$$

which we call the *structure operator* of the pseudocontext. Thus pseudocontexts are precisely pairs of distinct unit-norm rank-one decompositions of a single positive operator. This places them naturally among finite frame-theoretic constructions: contexts are the special case in which the structure operator is the identity on the relevant subspace; in general, pseudocontexts are unit-norm rank-one frame decompositions of a common positive operator. If the common operator is proportional to the identity on its support, the family is a tight frame; after normalisation on that support one obtains a POVM.

Remark 4 (Connection to SIC-POVMs). When the common structure operator satisfies $T = \frac{k}{n} I$ (tight frame on all of H), each family $\{P_{a_1},\dots,P_{a_k}\}$ and $\{P_{b_1},\dots,P_{b_k}\}$ scaled by n/k forms a symmetric informationally complete POVM (SIC-POVM) if $k = n^2$ and the frame vectors are equiangular [3]. Pseudocontexts therefore generalise the SIC-POVM notion by dropping both the tightness and equiangularity requirements.

IV. THE SMALLEST CASES

A. No pseudocontexts of size $k = 1$

Proposition 5 (Size $k = 1$)**.** *There is no pseudocontext of size* $k = 1$.

Proof. The condition $P(a_1) = P(b_1)$ for all $P \in S(L)$ means that the two self-adjoint operators P_{a_1} and P_{b_1} have equal expectation in every density operator. In finite dimension, density operators separate self-adjoint operators, so $P_{a_1} = P_{b_1}$. This rank-one identity forces a_1 and b_1 to span the same ray, contradicting the distinctness condition (ii) of Definition 2. □

B. Pseudocontexts of size $k = 2$ and the Bloch representation

The $k = 2$ case requires *complex* scalars; the proof that no genuine *real* pseudocontext of size $k = 2$ exists is given as Theorem 10(ii) below.

In dimension two the pseudocontext condition admits a complete characterisation in terms of Bloch vectors. This yields a unified and basis-independent description of all pseudocontexts and clarifies the real–complex dichotomy.

Every rank-one projector on $\mathbb{C}^2$ can be written as

$$P_a = \tfrac{1}{2}\big(I + \vec{r}_a \cdot \vec{\sigma}\big), \qquad \vec{r}_a \in \mathbb{R}^3, \quad \|\vec{r}_a\| = 1, \tag{5}$$

where $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices. To be specific, for any unit vector $a = \alpha e_1 + \beta e_2 \in \mathbb{C}^2$ in a chosen orthonormal basis (e_1, e_2), the components of the associated Bloch vector $\vec{r}_a = (r_x, r_y, r_z)$ are obtained by computing the expectation values of the Pauli matrices:

$$\begin{aligned} r_x &= \langle a, \sigma_x a\rangle = 2\,\mathrm{Re}(\alpha^*\beta), \\ r_y &= \langle a, \sigma_y a\rangle = 2\,\mathrm{Im}(\alpha^*\beta), \\ r_z &= \langle a, \sigma_z a\rangle = |\alpha|^2 - |\beta|^2. \end{aligned} \tag{6}$$

Proposition 6 (Bloch representation of pseudocontexts in dimension $d = 2$)**.** *Let $(a_1,\dots,a_k)$ and $(b_1,\dots,b_k)$ be k-tuples of unit vectors in $\mathbb{C}^2$ with Bloch vectors $\vec{r}_{a_i}, \vec{r}_{b_i} \in \mathbb{R}^3$. Then the following are equivalent:*

1. *$(a_1,\dots,a_k)$ and $(b_1,\dots,b_k)$ form a pseudocontext;*
2. $\sum_{i=1}^{k} P_{a_i} = \sum_{i=1}^{k} P_{b_i}$;
3. $\sum_{i=1}^{k} \vec{r}_{a_i} = \sum_{i=1}^{k} \vec{r}_{b_i}$.

The common structure operator T has Bloch form

$$T = \tfrac{k}{2} I + \tfrac{1}{2} \vec{R} \cdot \vec{\sigma}, \qquad \vec{R} = \sum_{i=1}^{k} \vec{r}_{a_i} = \sum_{i=1}^{k} \vec{r}_{b_i}.$$

Proof. By Definition 2 and (4), conditions 1 and 2 are equivalent. For the equivalence with 3, substitute the Bloch representation (5) into the structure operator:

$$\sum_{i=1}^{k} P_{a_i} = \sum_{i=1}^{k} \tfrac{1}{2}\big(I + \vec{r}_{a_i} \cdot \vec{\sigma}\big) = \tfrac{k}{2} I + \tfrac{1}{2}\Big(\sum_{i=1}^{k} \vec{r}_{a_i}\Big) \cdot \vec{\sigma}.$$

The identity component $\frac{k}{2} I$ is the same for both tuples, so equality of the two structure operators reduces to equality of the vector sums, giving the stated Bloch form of T. □

C. Explicit $k=2$ example via Bloch vectors

We can now construct an explicit $k=2$ pseudocontext by directly applying the Bloch representation and showing exactly how it arises from non-orthogonal configurations.

Proposition 7 (Existence of $k=2$ complex pseudocontexts)**.** *There are genuinely nonorthogonal pseudocontexts of size $k=2$ in $\mathbb{C}^n$ for every $n\geq 2$.*

Proof. Embed $\mathbb{C}^2$ into $\mathbb{C}^n$ as the span of the first two basis vectors e_1,e_2. Define two pairs of unit vectors:

$$u_\pm=\left(\tfrac{\sqrt{3}}{2},\pm\tfrac{1}{2},0,\dots,0\right),\qquad v_\pm=\left(\tfrac{\sqrt{3}}{2},\pm\tfrac{i}{2},0,\dots,0\right).$$

We obtain their Bloch vectors explicitly using (6). For $u_\pm$, setting $\alpha=\sqrt{3}/2$ and $\beta=\pm 1/2$, we compute:

$$\begin{aligned}r_x&=2\,\mathrm{Re}\big(\tfrac{\sqrt{3}}{2}\cdot(\pm\tfrac{1}{2})\big)=\pm\tfrac{\sqrt{3}}{2},\\ r_y&=2\,\mathrm{Im}\big(\tfrac{\sqrt{3}}{2}\cdot(\pm\tfrac{1}{2})\big)=0,\\ r_z&=\big|\tfrac{\sqrt{3}}{2}\big|^2-\big|\pm\tfrac{1}{2}\big|^2=\tfrac{3}{4}-\tfrac{1}{4}=\tfrac{1}{2}.\end{aligned}$$

Thus $\vec r_{u_\pm}=(\pm\frac{\sqrt{3}}{2},0,\frac{1}{2})$, which lie in the xz-plane of the Bloch sphere.

For $v_\pm$, setting $\alpha=\sqrt{3}/2$ and $\beta=\pm i/2$, we compute:

$$\begin{aligned}r_x&=2\,\mathrm{Re}\big(\tfrac{\sqrt{3}}{2}\cdot(\mp\tfrac{i}{2})\big)=0,\\ r_y&=2\,\mathrm{Im}\big(\tfrac{\sqrt{3}}{2}\cdot(\mp\tfrac{i}{2})\big)=\pm\tfrac{\sqrt{3}}{2},\\ r_z&=\big|\tfrac{\sqrt{3}}{2}\big|^2-\big|\pm\tfrac{i}{2}\big|^2=\tfrac{3}{4}-\tfrac{1}{4}=\tfrac{1}{2}.\end{aligned}$$

Thus $\vec r_{v_\pm}=(0,\pm\frac{\sqrt{3}}{2},\frac{1}{2})$, which lie in the yz-plane.

Summing the Bloch vectors within each pair yields the same structure vector:

$$\vec R=\vec r_{u_+}+\vec r_{u_-}=\vec r_{v_+}+\vec r_{v_-}=(0,0,1).$$

By Proposition 6, this guarantees that the operator sums match in the two-dimensional subspace. Embedding back into $\mathbb{C}^n$ gives, with $V:=\mathrm{span}\{e_1,e_2\}$,

$$\begin{aligned}P_{u_+}+P_{u_-}&=P_{v_+}+P_{v_-}=\tfrac{3}{2}P_{e_1}+\tfrac{1}{2}P_{e_2}\\ &=\big(I_V+\tfrac{1}{2}\sigma_z\big)\oplus 0_{n-2}=\begin{pmatrix}3/2&0\\0&1/2\end{pmatrix}\oplus 0_{n-2}.\end{aligned}$$

The four rays are distinct: u_+ and u_- differ by a sign in the real second coordinate; v_+ and v_- differ by a sign in the imaginary second coordinate; and $u_\pm$ versus $v_\pm$ differ by a factor of i in the second coordinate, which is not a real scalar. Within each pair, $\langle u_+,u_-\rangle=\frac{1}{2}\neq 0$ and $\langle v_+,v_-\rangle=\frac{1}{2}\neq 0$, so neither pair is orthogonal. The pseudocontext is therefore genuine. ☐

Remark 8 (Bloch geometry of the complex $k=2$ example). This construction directly illustrates the continuous $U(1)$ freedom present in complex spaces. The pair of Bloch vectors $\{\vec r_{v_+},\vec r_{v_-}\}$ is related to $\{\vec r_{u_+},\vec r_{u_-}\}$ by a 90° rotation about the z-axis (which aligns with their common sum $\vec R$). Because this rotation preserves $\vec R$, it produces a distinct, valid decomposition of the same fixed structure operator T.

D. Classification in $\mathbb{C}^2$

Pseudocontexts in dimension two correspond exactly to distinct decompositions of a fixed vector $\vec R\in\mathbb{R}^3$ into k unit vectors.

Theorem 9 (Complete characterisation in $\mathbb{C}^2$)**.** *Let $k\geq 1$.*

(i) For $k=1$, no pseudocontext exists.

(ii) For $k=2$, pseudocontexts exist and form continuous families.

(iii) For all $k\geq 3$, pseudocontexts exist; if $\vec R=0$, they arise from zero-centroid configurations (e.g. regular polygons).

Proof. Part (i) is Proposition 5.

For (ii), fix $\vec R\in\mathbb{R}^3$ with $0<\|\vec R\|<2$. The set of pairs $\{\vec r_1,\vec r_2\}$ of unit vectors satisfying $\vec r_1+\vec r_2=\vec R$ is parametrised by a circle: writing $\vec r_1=\vec R/2+\vec w$ and $\vec r_2=\vec R/2-\vec w$, the unit-norm condition $\|\vec r_i\|=1$ becomes $\|\vec w\|^2=1-\|\vec R\|^2/4$ together with $\vec w\perp\vec R$, so $\vec w$ ranges over a circle of radius $\sqrt{1-\|\vec R\|^2/4}$ in the plane perpendicular to $\vec R$ through the origin. Distinct positions $\vec w\neq\pm\vec w'$ on this circle yield distinct unordered pairs $\{\vec r_1,\vec r_2\}\neq\{\vec r_1',\vec r_2'\}$, and hence (by Proposition 6) two-tuples that form a pseudocontext.

For (iii), take $\vec R=0$. In $\mathbb{R}^3$ there are infinitely many distinct triples of unit vectors with zero sum, e.g. the vertices of any regular triangle centred at the origin, or any rotated version thereof. By Proposition 6, any two such distinct triples yield a pseudocontext of size $k=3$. ☐

E. Real–complex dichotomy in two dimensions

Theorem 10 (Real vs. complex in $d=2$)**.**

(i) In $\mathbb{C}^2$, genuine pseudocontexts of size $k=2$ exist.

(ii) In $\mathbb{R}^2$, no genuine pseudocontext of size $k=2$ exists.

(iii) In $\mathbb{R}^2$, genuine pseudocontexts exist for all $k\geq 3$.

Proof. In $\mathbb{C}^2$, rotations about a fixed Bloch vector $\vec R\neq 0$ form a continuous $U(1)$ symmetry, yielding inequivalent decompositions; this is the content of Theorem 9(ii).

In $\mathbb{R}^2$, Bloch vectors lie on a circle. For $k=2$, the condition $\vec r_1+\vec r_2=\vec R$ uniquely determines the pair up to permutation, so no second decomposition exists. If $\vec R=0$, the vectors are orthogonal and form a context.

For $k\geq 3$, distinct planar configurations with identical vector sum exist (e.g. rotated regular polygons). ☐

Remark 11 (Earlier observation). The nonexistence of genuine real pseudocontexts of size $k=2$ in dimension two was observed in [4] without noticing that the situation is different in complex spaces. The Bloch picture above makes the real–complex dichotomy transparent: over $\mathbb{R}$ a decomposition of a fixed structure operator into two rank-one terms is unique up to permutation, whereas over $\mathbb{C}$ a continuous phase degree of freedom produces inequivalent decompositions.

F. Spectral form

Proposition 12 (Spectrum in $d=2$). *The eigenvalues of T are*

$$\lambda_{\pm} = \tfrac{k}{2} \pm \tfrac{1}{2}\|\vec{R}\|.$$

Remark 13. The vector $\vec{R}$ completely parametrises pseudocontexts in dimension two.

V. PSEUDOCONTEXTS OF EVERY SIZE $k \geq 2$

The $k=2$ case is best understood via the Bloch-sphere parametrisation; the general k-construction below is its root-of-unity extension. The construction below uses a continuous phase $e^{i\phi}$ and is therefore intrinsically complex; the corresponding existence statement over $\mathbb{R}$ for $k \geq 3$ is covered separately by Example 16 in Section VI.

Theorem 14 (Complex-phase construction). *Let H contain an orthonormal pair e_1, e_2. Fix an integer $k \geq 2$, let $\omega = e^{2\pi i/k}$, choose $\alpha \in (0, \pi/2)$, and choose $\phi \in \mathbb{R}$ such that $e^{i\phi}$ is not a kth root of unity. Define unit vectors*

$$\begin{aligned} a_j &= \cos\alpha\, e_1 + \sin\alpha\, \omega^j e_2, \\ b_j &= \cos\alpha\, e_1 + \sin\alpha\, e^{i\phi}\omega^j e_2, \qquad j = 0, \ldots, k-1. \end{aligned}$$

Then the $2k$ rays spanned by the a_j and b_j are pairwise distinct, and

$$\sum_{j=0}^{k-1} P_{a_j} = \sum_{j=0}^{k-1} P_{b_j} = k\big(\cos^2\alpha\, P_{e_1} + \sin^2\alpha\, P_{e_2}\big).$$

Hence $(a_0, \ldots, a_{k-1})$ and $(b_0, \ldots, b_{k-1})$ form a pseudocontext of size k. If $\alpha \neq \pi/4$, then the pseudocontext is genuine.

Proof. Write $c = \cos\alpha$ and $s = \sin\alpha$. For each j,

$$\begin{aligned} P_{a_j} &= c^2 P_{e_1} + s^2 P_{e_2} + cs\big(\omega^{-j}|e_1\rangle\langle e_2| + \omega^j |e_2\rangle\langle e_1|\big), \\ P_{b_j} &= c^2 P_{e_1} + s^2 P_{e_2} + cs\big(e^{-i\phi}\omega^{-j}|e_1\rangle\langle e_2| + e^{i\phi}\omega^j |e_2\rangle\langle e_1|\big). \end{aligned}$$

Summing over $j = 0, \ldots, k-1$ and using $\sum_{j=0}^{k-1}\omega^j = 0$ cancels all off-diagonal terms, giving

$$\sum_{j=0}^{k-1} P_{a_j} = \sum_{j=0}^{k-1} P_{b_j} = k\big(c^2 P_{e_1} + s^2 P_{e_2}\big).$$

Distinctness. If $a_j = \lambda a_\ell$ for some scalar λ, then comparing first coordinates (which equal $c \neq 0$) gives $\lambda = 1$, and then $\omega^j = \omega^\ell$, so $j = \ell$. Hence the a_j are pairwise distinct rays; the identical argument applies to the b_j. If $a_j = \lambda b_\ell$, then $\lambda = 1$ and $\omega^j = e^{i\phi}\omega^\ell$, i.e. $e^{i\phi} = \omega^{j-\ell}$, contradicting the assumption that $e^{i\phi}$ is not a kth root of unity. Hence no a_j coincides with any b_ℓ.

Genuineness. For $j \neq \ell$ within the a-tuple,

$$\langle a_j, a_\ell\rangle = c^2 + s^2\omega^{\ell-j}.$$

This vanishes only if $c^2 = s^2$ (i.e. $\alpha = \pi/4$) and $\omega^{\ell-j} = -1$. So for $\alpha \neq \pi/4$, no two distinct vectors in the a-tuple are orthogonal. An identical computation gives $\langle b_j, b_\ell\rangle = c^2 + s^2\omega^{\ell-j}$, so the same conclusion holds for the b-tuple. Hence neither tuple is a context when $\alpha \neq \pi/4$. ☐

Remark 15. At the special value $\alpha = \pi/4$, the structure operator becomes $\frac{k}{2} I_{\mathrm{span}\{e_1, e_2\}}$, so the construction yields a unit-norm tight frame on the two-dimensional support. For $k=2$ this reduces to a context; for $k \geq 3$ it gives a highly symmetric pseudocontext (cf. Remark 17). In dimension $d=2$, this construction reduces to the Bloch-vector description of Sec. IV B.

VI. A SYMMETRIC $k=3$ EXAMPLE

The case $k=3$ admits a very transparent planar representation; it is the finite-dimensional analogue of the Mercedes–Benz frame.

Example 16 (Mercedes–Benz type pseudocontext). Fix a two-dimensional subspace $V \subset H$ with orthonormal basis (e_1, e_2), and fix an angle θ satisfying $\theta \not\equiv 0$, $\theta \not\equiv \pi/3$, and $\theta \not\equiv 2\pi/3$ modulo π. Put, for $j = 0, 1, 2$,

$$a_j = \cos\!\Big(\tfrac{2\pi j}{3}\Big) e_1 + \sin\!\Big(\tfrac{2\pi j}{3}\Big) e_2,$$

$$b_j = \cos\!\Big(\tfrac{2\pi j}{3} + \theta\Big) e_1 + \sin\!\Big(\tfrac{2\pi j}{3} + \theta\Big) e_2.$$

Then the six rays are pairwise distinct, and

$$\sum_{j=0}^{2} P_{a_j} = \sum_{j=0}^{2} P_{b_j} = \tfrac{3}{2} I_V.$$

Thus (a_0, a_1, a_2) and (b_0, b_1, b_2) form a genuine pseudocontext of size $k=3$ over $\mathbb{R}$.

Proof. In the basis (e_1, e_2), each projector has the form

$$P_{\cos\varphi\, e_1 + \sin\varphi\, e_2} = \begin{pmatrix} \cos^2\varphi & \cos\varphi\sin\varphi \\ \cos\varphi\sin\varphi & \sin^2\varphi \end{pmatrix}.$$

For $\varphi_j = 2\pi j/3$ and for $\varphi_j = \theta + 2\pi j/3$, the trigonometric identities

$$\sum_{j=0}^{2}\cos(2\varphi_j) = 0, \qquad \sum_{j=0}^{2}\sin(2\varphi_j) = 0,$$

which hold for any arithmetic progression of three angles spaced $2\pi/3$ apart, imply that the off-diagonal terms cancel while the diagonal terms each sum to $3/2$. Hence both sums equal $\frac{3}{2} I_V$.

The three excluded residue classes modulo π account for all collisions between the two triples:

- $\theta \equiv 0 \pmod{\pi}$: every b_j coincides with a_j or $-a_j$, giving the same ray.

- $\theta \equiv \pi/3 \pmod{\pi}$: the b-triple is a cyclic permutation of the a-triple as a set of rays.
- $\theta \equiv 2\pi/3 \pmod{\pi}$: the b-triple is again a permutation of the a-triple.

For all other values of θ, the six rays are pairwise distinct. Genuineness holds because for generic θ no two vectors within the same triple are orthogonal (the angle between consecutive vectors in each triple is $2\pi/3 \neq \pi/2$), so neither triple is a context. □

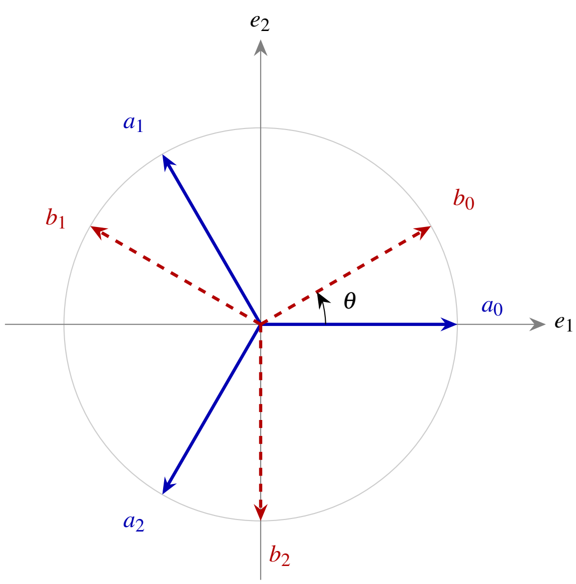

FIG. 1. The Mercedes–Benz pseudocontext of Example 16, drawn for $\theta = \pi/6$. The solid blue rays a_0, a_1, a_2 and the dashed red rays b_0, b_1, b_2 are two distinct unordered triples in the plane V, yet the associated rank-one projectors have the same sum $\frac{3}{2} I_V$.

Remark 17 (Complex trine description). Example 16 has a compact complex analogue. Let $\omega = e^{2\pi i/3}$ and embed V into $\mathbb{C}^n$ as $\mathrm{span}\{e_1, e_2\}$. The *trines*

$$u_j = \tfrac{1}{\sqrt{2}}\big(1, \omega^j, 0, \ldots, 0\big),\; v_j = \tfrac{1}{\sqrt{2}}\big(1, e^{i\phi}\omega^j, 0, \ldots, 0\big),$$

with $j = 0, 1, 2$, yield another rank-one decomposition of the same structure operator $\frac{3}{2} I_V$ for any $\phi \in \mathbb{R}$ that is not a multiple of $2\pi/3$. The identity $1 + \omega + \omega^2 = 0$ makes the off-diagonal entries of $\sum_j P_{u_j}$ and $\sum_j P_{v_j}$ cancel, giving the common structure operator $\frac{3}{2} I_V$. This form is most natural in frame-theoretic discussions and in connection with the Weyl–Heisenberg group.

Remark 18 (Minimal ambient dimension). Every pseudocontext of size $k \geq 2$ can be realised inside a two-dimensional subspace, and hence embedded into $\mathbb{C}^n$ for any $n \geq 2$: all constructions in Sections IV B and V live in $\mathrm{span}\{e_1, e_2\}$. For $k \geq 3$ the construction also works over $\mathbb{R}^n$. No pseudocontext of any size exists in dimension $n = 1$, since there every unit vector spans the unique ray, making the distinctness condition $|\{a_i\} \cup \{b_i\}| = 2k$ impossible.

Remark 19 (Nondegenerate structure operators). The trine family is highly symmetric: its structure operator is a scalar multiple of the identity on its support. More complicated examples with nondegenerate structure operators occur already in dimension three. The main difference is that the pseudocontext in Example 16 has only one constant value in the probability sum, while the advanced constructions admit a larger range of values, still respecting the equality. Explicit three-dimensional implementations of nondegenerate structure operators were derived in [5].

- **15-atom hypergraph (T_1).** A pseudocontext of size $k = 3$ yields $\lambda_1 = 2$, $\lambda_2 = \frac{7+\sqrt{21}}{14} \approx 0.827$, $\lambda_3 = \frac{7-\sqrt{21}}{14} \approx 0.173$, with $\mathrm{tr}(T_1) = 3$ and $\lambda_{\max} = 2 > 1$, consistent with Proposition 24.
- **36-atom hypergraph (T_2).** A larger true-implies-false gadget gives $T_2 = \mathrm{diag}(1/5, 1/5, 13/5)$ with $\mathrm{tr}(T_2) = 3$ and $\lambda_{\max} = 13/5 > 1$.

Such examples are useful when one wants to move beyond purely symmetric frame orbits.

VII. SPECTRAL STRUCTURE AND CANONICAL FORMS

In dimension $d = 2$, all spectral properties reduce to the Bloch form of Sec. IV B. We now analyse the general properties of the structure operator $T = \sum_{i=1}^{k} P_{a_i}$ associated with a k-tuple of unit vectors $(a_1, \ldots, a_k)$ in H.

A. Gram representation and spectral equivalence

Definition 20 (Analysis operator and Gram matrix)**.** Let $A\colon \mathbb{C}^k \to H$ be defined by $Ae_i = a_i$, where (e_i) is the standard basis of $\mathbb{C}^k$. The *analysis operator* is $A^\dagger\colon H \to \mathbb{C}^k$, and the *Gram matrix* is

$$G := A^\dagger A = \big(\langle a_i, a_j\rangle\big)_{i,j=1}^{k}.$$

Proposition 21 (Frame operator factorisation)**.** *The structure operator admits the factorisation $T = AA^\dagger$, and $G = A^\dagger A$.*

Proposition 22 (Spectral equivalence)**.** *The nonzero spectra of T and G coincide (including multiplicities):*

$$\sigma(T) \setminus \{0\} = \sigma(G) \setminus \{0\}.$$

In particular, $\mathrm{rank}(T) = \mathrm{rank}(G) \leq \min(k, n)$.

Proof. This is the standard singular-value correspondence between $AA^\dagger$ and $A^\dagger A$: if $AA^\dagger v = \lambda v$ with $\lambda \neq 0$, then $A^\dagger A(A^\dagger v) = \lambda (A^\dagger v)$ and $A^\dagger v \neq 0$. □

B. Spectral constraints

Proposition 23 (Trace identities)**.** *The structure operator satisfies*

$$\mathrm{tr}(T) = k, \qquad \mathrm{tr}(T^2) = \sum_{i,j=1}^{k} |\langle a_i, a_j\rangle|^2.$$

The diagonal terms $i=j$ each contribute 1*, giving* $\mathrm{tr}(T^2)\geq k$*, with equality if and only if the vectors are mutually orthogonal.*

Proof. $\mathrm{tr}(T)=\sum_i \mathrm{tr}(P_{a_i})=k$. For the second identity, $\mathrm{tr}(T^2)=\sum_{i,j}\mathrm{tr}(P_{a_i}P_{a_j})=\sum_{i,j}|\langle a_i,a_j\rangle|^2$. Diagonal terms contribute k; off-diagonal terms are non-negative, and vanish simultaneously if and only if all off-diagonal inner products are zero. □

Proposition 24 (Non-orthogonality and spectral spread)**.** *If the vectors (a_i) are not mutually orthogonal, then* $\mathrm{tr}(T^2)>k$*, the spectrum of T is non-flat, and $\lambda_{\max}(T)>1$.*

Proposition 25 (Majorisation constraint)**.** *Suppose $k\leq n$. Let $\lambda(T)=(\lambda_1,\dots,\lambda_n)$ be the eigenvalues of T in decreasing order (padded with zeros to length n), and let $d=(1,\dots,1,0,\dots,0)\in\mathbb{R}^n$ have k ones and $n-k$ zeros. Then $\lambda(T)\succ d$.*

Proof. The diagonal entries of G are $G_{ii}=\|a_i\|^2=1$, so the diagonal of G is $(1,\dots,1)\in\mathbb{R}^k$. By the Schur–Horn theorem, $\lambda(G)\succ(1,\dots,1)$. Since G and T share the same nonzero eigenvalues (Proposition 22), padding $\lambda(G)$ with $n-k$ zeros gives $\lambda(T)$, and $\lambda(T)\succ d$ follows.

Overcomplete case $k>n$. When $k>n$ the Gram matrix G is $k\times k$ with rank at most n; the majorisation is then best stated for the k-dimensional eigenvalue vector of G against $(1,\dots,1)\in\mathbb{R}^k$, and the Schur–Horn argument applies to G directly. □

Proposition 26 (Operator bounds)**.** *The structure operator satisfies $0\leq T\leq kI$. Moreover, if $\mu:=\max_{i\neq j}|\langle a_i,a_j\rangle|$, then*

$$\lambda_{\max}(T)\leq 1+(k-1)\mu.$$

Proof. Positivity $T\geq 0$ is immediate, and $T\leq kI$ since each $P_{a_i}\leq I$. For the eigenvalue bound, apply the Geršgorin circle theorem to G: since $G_{ii}=1$ and $|G_{ij}|\leq\mu$ for $i\neq j$, every eigenvalue of G lies in a disk centred at 1 with radius at most $(k-1)\mu$. Since the nonzero eigenvalues of T and G coincide, $\lambda_{\max}(T)\leq 1+(k-1)\mu$. □

C. Commutant and symmetry

Definition 27 (Commutant)**.** The *unitary commutant* of T is $\mathscr{C}(T):=\{U\in U(H):UT=TU\}$.

Proposition 28 (Structure of the commutant)**.** *Let $T=\sum_\alpha\lambda_\alpha P_\alpha$ be the spectral decomposition of T with distinct eigenvalues λ_α and orthogonal projections P_α of ranks m_α. Then $\mathscr{C}(T)\cong\prod_\alpha U(m_\alpha)$.*

Remark 29. Spectral degeneracies ($m_\alpha>1$) give rise to continuous families of distinct rank-one decompositions of T (as in tight frames), whereas a simple spectrum leads to strong rigidity. This explains why the tight-frame case (Remark 15) is so rich in pseudocontexts.

D. Canonical normal form

Proposition 30 (Spectral normal form)**.** *Every structure operator is unitarily equivalent to* $\mathrm{diag}(\lambda_1,\dots,\lambda_r,0,\dots,0)$ *with $\lambda_i>0$, $\sum_{i=1}^r\lambda_i=k$, and $r=\mathrm{rank}(T)$.*

E. Canonical parametrisation of all decompositions

Theorem 31 (Canonical decomposition)**.** *Let T be a positive operator of rank r with spectral decomposition $T=\sum_{i=1}^r\lambda_i|e_i\rangle\langle e_i|$, $\lambda_i>0$, and set $\Lambda=\mathrm{diag}(\lambda_1,\dots,\lambda_r)$. Assume $k\geq r$ (this is necessary, since $\mathrm{rank}(CC^\dagger)\leq k$).*

Every rank-one decomposition $T=\sum_{j=1}^k P_{a_j}$ with $\|a_j\|=1$ corresponds to a matrix $C\in\mathbb{C}^{r\times k}$ satisfying

$$CC^\dagger=\Lambda,\qquad \|c_j\|^2=1\ \text{for all}\ j, \tag{7}$$

where c_j is the j-th column of C, via $a_j=\sum_{i=1}^r c_{ij}e_i$. Conversely, every such C defines a valid decomposition.

Proof. Let $T=AA^\dagger$ with columns a_j. In the eigenbasis of T, write $A=UC$ where U is the unitary embedding the eigenbasis of T on its support into H. Then $CC^\dagger=U^\dagger TU\big|_{\mathrm{supp}(T)}=\Lambda$, and $\|c_j\|=\|a_j\|=1$. The converse is a direct computation. □

Corollary 32 (Unitary freedom)**.** *Among all rank-one decompositions $T=\sum_{j=1}^k|a_j\rangle\langle a_j|$ into k* not necessarily *unit-norm vectors, any two are related by $A\mapsto AU$ for some $U\in U(k)$ (this is the standard $U(k)$-orbit on the affine fibre $\{A:AA^\dagger=T\}$). The unit-norm constraint $\|c_j\|=1$ in (7) cuts out a real-codimension-k subvariety of this orbit, and what is relevant for pseudocontexts is the quotient by phases acting on individual columns and by permutations of columns, since rays—not vectors—are the data.*

F. Characterisation of pseudocontexts

Proposition 33 (Operator equivalence criterion)**.** *Two k-tuples $(a_1,\dots,a_k)$ and $(b_1,\dots,b_k)$ of distinct rays form a pseudocontext if and only if their matrices C_A,C_B in the eigenbasis of T both satisfy (7):*

$$C_AC_A^\dagger=C_BC_B^\dagger=\Lambda.$$

Remark 34. Pseudocontexts correspond to distinct points in the space $\{C\in\mathbb{C}^{r\times k}:CC^\dagger=\Lambda,\|c_j\|=1\}/\sim$, where $\sim$ identifies decompositions that yield the same set of rays.

G. Extremal spectral bounds

Proposition 35 (Rayleigh bounds)**.** *For any density operator ρ,*

$$\lambda_{\min}(T)\leq\mathrm{tr}(\rho T)\leq\lambda_{\max}(T).$$

Both bounds are tight, achieved by pure states on the corresponding eigenvectors of T.

VIII. TOPOLOGICAL FORCING VS. VECTORIAL REPRESENTATION: THE PROBLEM OF "LOOSE ENDS"

The discrepancy between the purely geometric validity of the $k=2$ complex pseudocontext and the failure of its orthogonality hypergraph to enforce it highlights a crucial distinction in how pseudocontexts can arise.

One and the same configuration of observables—drawn as a Greechie-type orthogonality hypergraph—can admit very different types of probability assignments. These include classical probabilities (mediated by convex sums of two-valued $\{0,1\}$-states) as well as quantum probabilities (mediated by density operators via the Cauchy–Gleason theorem [6]). However, any valid probability assignment on the hypergraph, regardless of its origin, must satisfy the fundamental completeness condition: the sum of probabilities on any context must equal one (that is, $P(I)=1$).

For certain tightly intertwined hypergraphs, this completeness condition alone is sufficient to structurally force a pseudocontext identity. The proof technique, successfully utilized in [4] and applicable to configurations like the aforementioned octagon, relies on finding two distinct "coverings." By summing the completeness conditions of a specific subset of contexts containing the first pseudocontext, and comparing it to a second subset containing the second pseudocontext, one can algebraically cancel out all intermediate auxiliary vertices. When this topological cancellation is perfect, the equality of the pseudocontext probability sums becomes a rigid, state-independent logical necessity of the hypergraph itself, applying equally to classical and quantum probabilities.

However, this covering technique cannot be applied to the $k=2$ complex pseudocontext formed by $u_\pm$ and $v_\pm$ in Proposition 7. If we attempt to equate the probability sums of the u-pair and the v-pair using the contexts provided, we obtain:

$$P(u_+)+P(u_-)=2-2P(e_3)-\big(P(w_+)+P(w_-)\big), \tag{8}$$
$$P(v_+)+P(v_-)=2-2P(e_3)-\big(P(z_+)+P(z_-)\big). \tag{9}$$

Subtracting (9) from (8) yields:

$$\begin{aligned} &P(u_+)+P(u_-)-\big(P(v_+)+P(v_-)\big)\\ &=P(z_+)+P(z_-)-\big(P(w_+)+P(w_-)\big). \end{aligned} \tag{10}$$

For the pseudocontext identity to be forced purely by the graph, the right-hand side would need to be forced to zero. But $w_\pm$ and $z_\pm$ are "loose ends": they each belong to exactly one context in the hypergraph. Because they do not intertwine with any other contexts, their probability values are completely unconstrained relative to one another. There is no topological mechanism to force $P(w_+)+P(w_-)=P(z_+)+P(z_-)$.

It is precisely these loose ends that allow the classical $\{0,1\}$-valuation (which assigns 0 to $w_\pm$ and 1 to $z_\pm$) to comfortably violate the pseudocontext equality. The identity $P(u_+)+P(u_-)=P(v_+)+P(v_-)$ holds for these specific quantum observables solely due to the "miracle" of their specific complex vector representation, not due to the logical skeleton of their orthogonalities.

To achieve a structurally forced pseudocontext—one where the identity is dictated purely by the graph—we require tighter bounds and configurations without such loose ends. The vertices must intertwine in such a way that they form closed topological loops, enabling the exact algebraic cancellations utilized in the covering proofs. This motivates the search for, and the study of, more complex, higher-dimensional hypergraphs where pseudocontextuality is embedded directly into the orthomodular logic.

IX. THE OCTAGON PSEUDOCONTEXT VIA DISTINCT COVERINGS

We can rigorously demonstrate the pseudocontextual nature of a tightened—relative to the previous configuration involving $u_\pm$ and $v_\pm$—octagon hypergraph by employing a proof technique based on mutually distinct "coverings," a method we previously utilized in [4].

The hypergraph consists of 20 vertices and 12 contexts of size three. We label the 16 outer vertices along the perimeter as follows: let $v_0,v_1,\ldots,v_7$ be the midpoints of the eight octagon edges, and let $v_{01},v_{12},\ldots,v_{70}$ be the eight corners. The perimeter is formed by eight straight-line contexts:

$$E_i=\{v_{i-1,i},\ v_i,\ v_{i,i+1}\}\quad \text{for } i=0,\ldots,7 \pmod 8.$$

Notice that adjacent midpoints, such as v_1 and v_2, are not on the same context; rather, their respective perimeter contexts E_1 and E_2 intertwine at the shared corner vertex v_{12}.

The remaining four inner contexts connect opposite midpoints and pass through the four inner vertices a_1,a_2 and b_1,b_2:

$$\begin{aligned} D_0&=\{v_0,a_1,v_4\}, & D_2&=\{v_2,a_2,v_6\},\\ D_1&=\{v_1,b_2,v_5\}, & D_3&=\{v_3,b_1,v_7\}. \end{aligned}$$

We aim to prove that the sums of probabilities on the inner pairs match: $P(a_1)+P(a_2)=P(b_1)+P(b_2)$. To do so, we construct two distinct coverings of the hypergraph.

Covering A focuses on the first pseudocontext pair $\{a_1,a_2\}$, as depicted in Fig. 2(a). We construct it by taking the inner contexts D_0,D_2 along with four alternating perimeter edges:

$$A=\{D_0,D_2,E_1,E_3,E_5,E_7\}.$$

Summing the completeness condition over these six contexts yields a total probability of 6. Let us evaluate the multiplicity of each vertex covered by A:

- The inner vertices a_1,a_2 are covered exactly once.
- The midpoints v_0,v_4 and v_2,v_6 are covered exactly once (by D_0,D_2).
- The remaining midpoints v_1,v_3,v_5,v_7 are covered exactly once (by E_1,E_3,E_5,E_7).
- Each of the eight corners $v_{i,i+1}$ is covered exactly once because the selected alternating perimeter contexts perfectly span all corners without overlapping.

Thus, covering A perfectly partitions all 16 outer vertices, covering each exactly once:

$$P(a_1)+P(a_2)+\sum_{i=0}^{7}P(v_i)+\sum_{\text{corners}}P(v_{i,i+1})=6. \qquad (11)$$

Covering B focuses on the second pseudocontext pair $\{b_1,b_2\}$, depicted in Fig. 2(b). We construct it using the other two inner contexts D_1,D_3 alongside the complementary set of four alternating perimeter edges:

$$B=\{D_1,D_3,E_0,E_2,E_4,E_6\}.$$

Summing over these six contexts also yields 6. Evaluating the vertex multiplicities reveals:

- The inner vertices b_1,b_2 are covered exactly once.
- The midpoints v_1,v_5 and v_3,v_7 are covered exactly once (by D_1,D_3).
- The remaining midpoints v_0,v_2,v_4,v_6 are covered exactly once (by E_0,E_2,E_4,E_6).
- The selected perimeter contexts E_0,E_2,E_4,E_6 again perfectly span all eight corners exactly once.

Hence, Covering B covers the same set of 16 outer vertices, each exactly once:

$$P(b_1)+P(b_2)+\sum_{i=0}^{7}P(v_i)+\sum_{\text{corners}}P(v_{i,i+1})=6. \qquad (12)$$

Because Covering A and Covering B perfectly match in their multiplicities over all 16 outer perimeter vertices, we can directly equate (11) and (12). The perimeter terms completely cancel out, leaving:

$$P(a_1)+P(a_2)=P(b_1)+P(b_2).$$

This rigorously establishes the state-independent probabilistic equivalence between the two pairs.

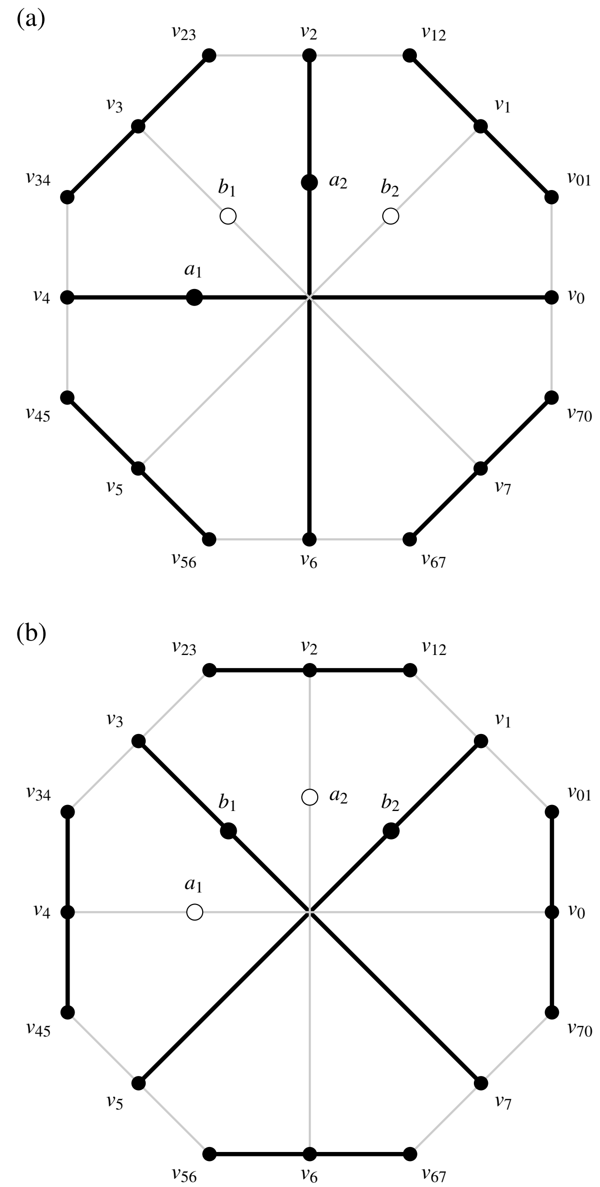

FIG. 2. Graphical representation of the two distinct coverings of the octagon hypergraph. **(a)** Covering A (bold lines) isolates the (a_1,a_2) pseudocontext using inner contexts D_0,D_2 and alternating outer edges. **(b)** Covering B (bold lines) isolates the (b_1,b_2) pseudocontext using inner contexts D_1,D_3 and the complementary outer edges. Vertices actively engaged in a covering are filled black, while the two strictly uncovered inner vertices in each panel are filled white. Since both coverings exactly partition all 16 outer vertices, their respective probability sums must equate, forcing $P(a_1)+P(a_2)=P(b_1)+P(b_2)$.

A. The parameter C

In the octagon coordinatization—that is, the search for a faithful orthogonal representation [7] of the uniform hypergraph [8] in which vectors assigned to adjacent vertices (or, more generally, to vertices within each hyperedge) are required to be orthogonal—it is convenient to parametrize the inner pair by a single positive parameter C. Since only rays matter, all vectors below are understood up to a nonzero scalar multiple.

Proposition 36 (Meaning of the parameter C). *Let $s>0$, set $C:=s^2$, and choose representatives*

$$a_1=(s,1,0), \qquad a_2=(s,-1,0),$$

with companion rays

$$b_1=(s,i,0), \qquad b_2=(s,-i,0).$$

Then the inner pair $\{a_1,a_2\}$ has mutual angle 2α satisfying

$$\cos(2\alpha)=\frac{C-1}{C+1}, \qquad \textit{equivalently} \qquad C=\cot^2\alpha.$$

In particular, $C=1$ is the orthogonal case.

Proof. The normalized representatives of a_1 and a_2 are

$$\frac{(s,1,0)}{\sqrt{s^2+1}}, \qquad \frac{(s,-1,0)}{\sqrt{s^2+1}}.$$

Their inner product equals $(s^2-1)/(s^2+1)$, which gives the stated formula after substituting $C=s^2$. The same computation applies to the b-pair. □

Remark 37. The point of the parameter C is that, in the explicit octagon coordinatization used in the main text, the closure conditions depend on the inner pair only through this single scalar.

B. Exact $\mathbb{C}^3$ coordinates at $C=1$

At $C=1$ the inner pair becomes orthogonal, and the numerical $\mathbb{C}^3$ solution found for the octagon admits a closed-form algebraic realization. As in the main text, the vectors below are ray representatives; normalization is not essential. The configuration is depicted in Fig. 3. A closer inspection shows that this configuration of 21 points in 14 contexts (hyperedges) supports a separating (and thus set representable [9]) set of 24 two-valued states.

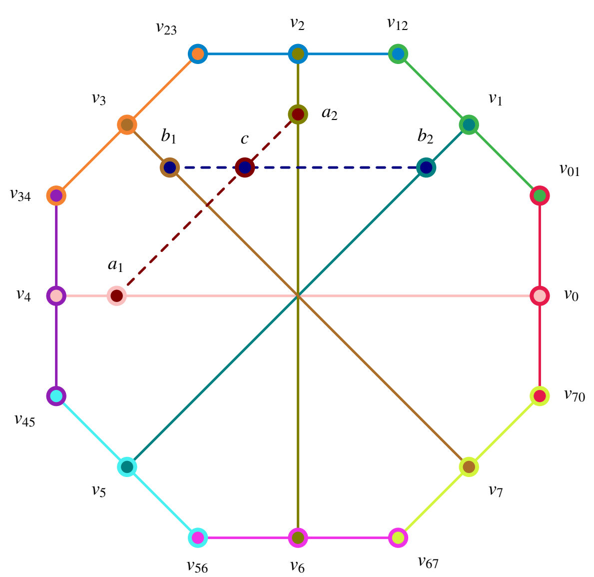

FIG. 3. Redraw of the octagon using exactly the same coordinates as Fig. 2(a)/(b). The 12 original contexts are drawn with solid lines in 12 distinct colors. The two additional $C=1$ contexts are dashed and meet at the intertwining vertex c. Concentric disks mark vertices incident with two contexts.

Example 38 (Exact $\mathbb{C}^3$ realization at $C=1$). Let

$$p=\frac{\sqrt{14}+\sqrt{2}}{4}, \qquad q=\frac{\sqrt{14}-\sqrt{2}}{4}.$$

Then

$$p^2+q^2=2, \qquad pq=\frac{3}{4}.$$

A convenient choice of the inner rays intersection at the additional point $c=(0,0,1)$ is

$$a_1=(1,1,0), \quad a_2=(1,-1,0),$$
$$b_1=(1,i,0), \quad b_2=(1,-i,0).$$

One compatible choice of the eight midpoint rays is

$$v_0=(1,-1,\sqrt{2}), \qquad v_4=(1,-1,-\sqrt{2}),$$
$$v_1=(1,i,p+iq), \qquad v_5=(1,i,-p-iq),$$
$$v_2=(1,1,-i\sqrt{2}), \qquad v_6=(1,1,i\sqrt{2}),$$
$$v_3=(1,-i,q-ip), \qquad v_7=(1,-i,-q+ip).$$

One compatible choice of the eight corner rays is

$$v_{01}=\big(-p+i(q+\sqrt{2}),\ \sqrt{2}-p+iq,\ 1-i\big),$$
$$v_{12}=\big(\sqrt{2}-p+iq,\ p-i(q+\sqrt{2}),\ 1+i\big),$$
$$v_{23}=\big(q+\sqrt{2}+ip,\ -q+i(\sqrt{2}-p),\ -1+i\big),$$
$$v_{34}=\big(q+i(p-\sqrt{2}),\ q+\sqrt{2}+ip,\ -1-i\big),$$
$$v_{45}=\big(p-i(q+\sqrt{2}),\ -\sqrt{2}+p-iq,\ 1-i\big),$$
$$v_{56}=\big(-\sqrt{2}+p-iq,\ -p+i(q+\sqrt{2}),\ 1+i\big),$$
$$v_{67}=\big(-q-\sqrt{2}-ip,\ q-i(\sqrt{2}-p),\ -1+i\big),$$
$$v_{70}=\big(-q-i(p-\sqrt{2}),\ -q-\sqrt{2}-ip,\ -1-i\big).$$

A direct symbolic check shows that each listed context is orthogonal. The identities $p^2+q^2=2$ and $pq=\frac{3}{4}$ are sufficient for the simplification. Hence the octagon from Section IX admits an exact $\mathbb{C}^3$ realization at $C=1$.

Remark 39. The coordinates above are not unique. Any common unitary transformation, together with arbitrary rescalings of the individual ray representatives, yields an equivalent realization.

Remark 40 (Phase symmetry). The exact $\mathbb{C}^3$ realization exploits a complex phase symmetry in the third coordinate. This is one of the reasons why the construction is much more rigid over $\mathbb{R}$ than over $\mathbb{C}$.

C. No faithful real realization at $C=1$

Example 38 provides an exact realization of the octagon hypergraph in $\mathbb{C}^3$ at the orthogonal limit $C=1$, where the inner pseudocontext splits into two contexts. We show that no faithful realization by real rays exists.

Proposition 41 (No faithful real realization at $C=1$)**.** *The octagon hypergraph of Section IX, augmented by the two inner $C=1$ contexts intersecting at the vertex c, admits no faithful representation by rays in $\mathbb{R}^3$.*

Proof. Assume a faithful real realization exists. By a global orthogonal transformation, fix $c=(0,0,1)^T$. All rays sharing a context with c lie in the xy-plane. Using rotational freedom in that plane, set

$$a_1=(1,0,0)^T, \qquad a_2=(0,1,0)^T.$$

Let

$$b_1 = (\cos\theta, \sin\theta, 0)^T, \qquad b_2 = (-\sin\theta, \cos\theta, 0)^T,$$

with $\theta \notin \frac{\pi}{2}\mathbb{Z}$ by faithfulness. Define $m = \tan\theta \neq 0$ and $M = 1 + m^2 > 1$.

Let the midpoint rays be $v_j = (X_j, Y_j, Z_j)^T$.

Nonplanarity of midpoint rays. If some $Z_j = 0$, then v_j lies in the xy-plane. If v_j belongs to a context with $a_1 = (1,0,0)^T$ (resp. a_2), orthogonality forces $X_j = 0$ (resp. $Y_j = 0$), leaving v_j proportional to a_2 (resp. a_1). Similarly, if v_j is constrained by b_1 or b_2, the relations $Y = mX$ or $X = -mY$ together with $Z_j = 0$ force coincidence with one of the b-rays. In all cases this violates faithfulness. Hence $Z_j \neq 0$ for all j, and we may normalize

$$v_j = (X_j, Y_j, 1)^T.$$

The inner contexts impose:

$$\begin{aligned}
D_0: &\quad X_0 = X_4 = 0, \qquad Y_0Y_4 + 1 = 0,\\
D_1: &\quad Y_1 = mX_1, \quad Y_5 = mX_5, \quad X_1X_5M + 1 = 0,\\
D_3: &\quad X_3 = -mY_3, \quad X_7 = -mY_7, \quad Y_3Y_7M + 1 = 0.
\end{aligned}$$

Thus

$$X_5 = -\frac{1}{MX_1}, \qquad Y_7 = -\frac{1}{MY_3}.$$

The quantities X_1 and Y_3 are nonzero: if $X_1 = 0$ (resp. $Y_3 = 0$), then v_1 (resp. v_3) becomes collinear with a coordinate direction, forcing vertex identifications incompatible with faithfulness.

For perimeter contexts, the condition that adjacent corner rays are orthogonal yields

$$(v_{j-1}\cdot v_j)(v_j \cdot v_{j+1}) - (v_{j-1}\cdot v_{j+1})\|v_j\|^2 = 0.$$

Evaluating at $j = 0$ and $j = 4$ gives

$$Y_0 = \frac{Y_7 + mX_1}{1 - mX_1Y_7}, \qquad Y_4 = \frac{Y_3 + mX_5}{1 - mY_3X_5}, \tag{13}$$

where the denominators are nonvanishing; otherwise the closure condition would enforce proportionality among distinct rays, contradicting faithfulness.

Substituting the inner relations into (13) and simplifying the complex fractions yields

$$Y_0 = \frac{mMX_1Y_3 - 1}{MY_3 + mX_1}, \qquad Y_4 = \frac{MX_1Y_3 - m}{MX_1 + mY_3}.$$

Using $Y_0Y_4 = -1$ from D_0, set $u = X_1Y_3$. Expanding the product of these fractions, collecting terms in u, and applying the identity $M = 1 + m^2$ causes all mixed terms to cancel, yielding the master constraint (see Appendix A)

$$(M^2u^2 + mu + 1) + M(X_1^2 + Y_3^2) = 0. \tag{14}$$

Since $X_1, Y_3 \in \mathbb{R}$,

$$X_1^2 + Y_3^2 \geq 2|X_1Y_3| = 2|u|,$$

and therefore the left-hand side is bounded below by

$$f(u) = M^2u^2 + mu + 1 + 2M|u|.$$

Using $M = 1 + m^2$, one has $2M - |m| > 0$, hence

$$M^2u^2 + mu + 2M|u| \geq M^2u^2 + |u|(2M - |m|) \geq 0,$$

with equality only at $u = 0$. Consequently $f(u) \geq 1$ for all $u \in \mathbb{R}$, contradicting the required vanishing. Therefore no real realization exists. □

Remark 42 (Complex realization). In the complex construction, one has formally $m = i$, hence $M = 1 + i^2 = 0$. The relations above then become singular, and the coercive quadratic structure responsible for the contradiction disappears. Substituting $M = 0$ reduces the constraint to $iu + 1 = 0$, which admits solutions in $\mathbb{C}$.

Remark 43. The paper [10] presented the first construction of an orthogonality (Greechie) diagram which can be represented by vectors in complex spaces, but not in real ones. As a byproduct of our effort, we found another (and much simpler) such diagram described in this section.

X. PHYSICAL INTERPRETATION AND CONSTRAINTS

A pseudocontext $(a_1, \ldots, a_k)$, $(b_1, \ldots, b_k)$ implements a state-independent empirical equivalence between two families of rank-one observables $A_i = P_{a_i}$ and $B_i = P_{b_i}$. Although the operators within each family are generally noncommuting, the two linear functionals

$$\rho \longmapsto \sum_i \mathrm{tr}(\rho A_i), \qquad \rho \longmapsto \sum_i \mathrm{tr}(\rho B_i)$$

agree on the whole state space.

The spectral decomposition of T establishes absolute quantum limits on these probability sums. By Proposition 35,

$$\lambda_{\min}(T) \leq \sum_{i=1}^{k} P(a_i) \leq \lambda_{\max}(T). \tag{15}$$

These bounds are tight.

For the 36-atom hypergraph example [5], the structure operator $T_2 = \mathrm{diag}(1/5, 1/5, 13/5)$ confines the quantum probability sum to $[1/5, 13/5] = [0.2, 2.6]$. A classical hidden-variable model based on two-valued ($\{0,1\}$) truth assignments to projections—subject to the consistency requirement that exactly one atom in each orthogonal basis receives the value 1—would allow the sum to range over $\{0,1,2,3\}$, which includes the extreme values 0 and 3. Because T_2 has no eigenvalue equal to 0 or 3, quantum mechanics forbids these extremes; the quantum system cannot reach them. This discrepancy is a signature of Kochen–Specker-type contextuality [9, 11].

This state-independent additive equivalence is related to, but distinct from, the operational notion of contextuality introduced by Spekkens [12], which applies to arbitrary experimental procedures (preparations, transformations, and unsharp measurements) rather than to sharp projective measurements alone. A precise comparison is beyond the scope of this note.

In the language of quantum information, $(A_1,\dots,A_k)$ and $(B_1,\dots,B_k)$ are distinct rank-one decompositions of the same positive operator. When this operator is proportional to the identity on its support, the families form tight frames; after normalisation they yield POVMs.

XI. OPEN PROBLEMS

Open Problem (Classification). Classify, up to unitary equivalence, all pseudocontexts of size k in $\mathbb{C}^n$. Even for small (k,n), the orbit structure is not yet transparent.

Open Problem (Lattice representability). Characterise which orthomodular lattices carrying a pseudocontext-type additive identification admit a faithful embedding into some $L(\mathbb{C}^n)$, sharpening the partial results of [5, 13].

Open Problem (Frame-symmetry characterisation). Can one characterise pseudocontexts purely in terms of symmetries of the underlying rank-one frame, for example in terms of unitaries commuting with the structure operator and permuting the atoms of the two decompositions?

Open Problem (Higher-dimensional pseudocontexts). Classify pseudocontexts that genuinely span a higher-dimensional subspace. In particular, in [5, 13] and here in Section IX, we studied systems of vectors such that their *orthogonality relations* themselves induce the satisfaction of condition (iii) of Definition 2 for any vector representation fulfilling the orthogonality constraints. The systematic classification of such higher-dimensional pseudocontexts remains open.

Open Problem (Classification of minimal real pseudocontexts). Theorem 10 establishes that genuine real pseudocontexts of size $k=2$ do not exist in any dimension, while Example 16 demonstrates their existence for $k=3$. Thus $k=3$ is the minimal size for genuine real pseudocontexts. Classify all such minimal ($k=3$) examples up to orthogonal equivalence over $\mathbb{R}$.

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Appendix A: Derivation of the Master Constraint

We provide a step-by-step verification of the algebraic derivation leading from the perimeter closure conditions to the master constraint (14) used in the proof of Proposition 41.

The starting point is the perimeter closure condition stated in the proof: for three rays v_{j-1}, v_j, v_{j+1} forming a context, adjacent corner rays are orthogonal, yielding

$$(v_{j-1}\cdot v_j)(v_j\cdot v_{j+1}) - (v_{j-1}\cdot v_{j+1})\|v_j\|^2 = 0. \tag{A1}$$

Evaluating (A1) at $j=0$ and $j=4$ produces the relations

$$Y_0 = \frac{Y_7 + mX_1}{1 - mX_1Y_7}, \tag{A2}$$

$$Y_4 = \frac{Y_3 + mX_5}{1 - mY_3X_5}, \tag{A3}$$

as stated in the proof. The inner contexts impose the constraints

$$Y_7 = -\frac{1}{MY_3}, \qquad X_5 = -\frac{1}{MX_1}, \qquad M = 1+m^2, \tag{A4}$$

together with the requirement $Y_0Y_4 = -1$ from context D_0.

Simplifying the fractions

Substituting Y_7 from (A4) into (A2):

$$Y_0 = \frac{-\frac{1}{MY_3} + mX_1}{1 - mX_1\left(-\frac{1}{MY_3}\right)} = \frac{mMX_1Y_3 - 1}{MY_3 + mX_1}. \tag{A5}$$

Substituting X_5 from (A4) into (A3):

$$Y_4 = \frac{Y_3 + m\left(-\frac{1}{MX_1}\right)}{1 - mY_3\left(-\frac{1}{MX_1}\right)} = \frac{MX_1Y_3 - m}{MX_1 + mY_3}. \tag{A6}$$

Constructing the product condition

Let $u = X_1Y_3$. Using the constraint $Y_0Y_4 = -1$ from context D_0 together with (A5) and (A6):

$$\frac{mMu-1}{mX_1+MY_3}\cdot\frac{Mu-m}{MX_1+mY_3} = -1. \tag{A7}$$

Cross-multiplying (A7) yields

$$(mMu-1)(Mu-m) = -(mX_1+MY_3)(MX_1+mY_3). \tag{A8}$$

Expanding both sides

The left-hand side of (A8) expands as

$$\begin{aligned}\text{LHS} &= mM^2u^2 - m^2Mu - Mu + m\\ &= mM^2u^2 - Mu(m^2+1) + m.\end{aligned} \tag{A9}$$

Since $M = 1+m^2$, the coefficient m^2+1 equals M, giving

$$\text{LHS} = mM^2u^2 - M^2u + m. \tag{A10}$$

The right-hand side of (A8) expands as

$$\begin{aligned}\text{RHS} &= -\left[mMX_1^2 + m^2X_1Y_3 + M^2X_1Y_3 + mMY_3^2\right]\\ &= -\left[u(M^2+m^2) + mM(X_1^2+Y_3^2)\right].\end{aligned} \tag{A11}$$

Cancellation and the master constraint

Equating (A10) and (A11):

$$mM^2u^2 - M^2u + m = -u(M^2 + m^2) - mM(X_1^2 + Y_3^2). \quad \text{(A12)}$$

Adding M^2u to both sides, the $-M^2u$ term cancels:

$$mM^2u^2 + m = -m^2u - mM(X_1^2 + Y_3^2). \quad \text{(A13)}$$

Bringing all terms to the left and dividing by $m \neq 0$ (since $\theta \notin \frac{\pi}{2}\mathbb{Z}$ by faithfulness) yields the master constraint (14) in the proof of Proposition 41.

Positivity contradiction

Since $X_1, Y_3 \in \mathbb{R}$, the inequality $X_1^2 + Y_3^2 \geq 2|X_1Y_3| = 2|u|$ holds. The left-hand side of (14) therefore satisfies

$$\begin{aligned} &M^2u^2 + mu + 1 + M(X_1^2 + Y_3^2) \\ &\geq M^2u^2 + mu + 1 + 2M|u| \\ &= M^2u^2 + |u|(2M - |m|) + 1. \end{aligned} \quad \text{(A14)}$$

With $M = 1 + m^2 > 1$, we have $2M - |m| = 2 + 2m^2 - |m| > 0$ for all real m. Consequently the expression in (A14) is strictly positive for every $u \in \mathbb{R}$, contradicting (14). This completes the algebraic verification of the master constraint and the resulting impossibility of a real realization. "'

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